Raghav Arora

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Summary: Robotics Engineer specializing in autonomous mobile manipulation and foundation model-guided task planning.

EDUCATION

M.S., Electrical and Computer Engineering | May 2026
The University of Texas at Austin │ GPA: 3.86
Relevant Coursework: Tactile Sensing for Robotics, Adv. Robot Manipulation and Learning, Generative Models for ML
B.E., Electrical and Electronics Engineering; M.Sc. Chemistry| May 2022
Birla Institute of Technology and Science, Pilani │ GPA: 8.86 (out of 10)

PROFESSIONAL EXPERIENCE

RoboCup 2025 | UT Austin

- Secured 2nd place in the RoboCupAtHome Domestic Standard Platform League in Salvador, Brazil.
- Architected a unified system for general-purpose service robots, ensuring robust performance across diverse lighting and high-density crowded environments.
- Engineered a tracking pipeline using Behavior Trees and Multi-Pedestrian Tracking to execute precise human-following in crowds of 100+ onlookers.

Graduate Research Assistant | April 2024 – Present
UT Austin (Supervisors: Prof. Peter Stone, Dr. Yoonchang Sung)
Project: Anticipatory Reinforcement Learning for Improving Continual Planning in homes.

- Devised a task conditioned Markov Decision Process that anticipates future tasks and optimizes anticipated tasks using Deep Reinforcement Learning.

Project: VLM Guided Task and Motion Planning

- Developed correlational particle filter integrating LLM-generated object pose beliefs with visual observations.
- Created 3D Scene Graph Representation for robot observations, and a framework for VLM-guided TAMP .
- Reduced real-world execution time on Toyota HSR by 67% vs. baseline planners.

Research Engineer | Oct 2022 – April 2024
IIIT Hyderabad
Project: Foundation models for Household Robotics

- Designed multimodal graph networks to enable commonsense reasoning in household robotics, surpassing baseline LLM success rates by 42%.
- Leveraged Large Language Models (LLMs) to extract and model user patterns, implementing predictive task anticipation logic that reduced task execution time by 31%.

Relevant Publications

- Large-Language-Model-Guided State Estimation for Partially Observable Task and Motion Planning: Y. Kim, **Raghav Arora**, R. Martín-Martín, P. Stone, B. Abbatematteo, Y. Sung
  In: IEEE International Conference on Robotics and Automation (ICRA) 2026. Vienna, Austria
- Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments: **Raghav Arora**, S. Singh, K. Swaminathan, A. Datta, B. Bhowmick, K. Jatavallabhula, M. Sridharan, M. Krishna
  In: IEEE International Conference on Robotics and Automation (ICRA) 2024. Yokohama, Japan

Technical Skills

- **Areas of Interest**: *Robotics, Mobile Manipulation, TAMP, Perception, Generative AI, Reinforcement Learning*
- **Proficient in:** *Python, Pytorch, Keras, Git, LaTeX, C++, Java, Docker, ROS/ROS2*
- **Robotic Simulators:** *IsaacLab,Mujoco, Pybullet, CoppeliaSim, AI2THOR, AIHabitat*
- **Robotic Platforms:** *Toyota HSR, Booster T1 (humanoid), Cobot Mobile Manipulator, and custom robots*